Torsion in Magnitude Homology of Graphs

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Introduction

In 2013 Leinster developed the theory of the magnitude of a finite metric space [4], and in the following year speacilized magnitude to become a graph invariant[5]. Magnitude homology of graphs is the categorification of this invariant[2]. Then in 2018 Kaneta and Yoshinaga found a graph with torsion[3], and in 2021 Sazdanovic and Summers were able to generalize this approach[6]. Here we ask and answer the question, "Can a 'smaller' graph (in terms of vertices and edges) than the one presented first by the Kaneta-Yoshinaga Sazdanovic-Summers approach be found such that it has torsion?"

Background

For a graph $G=(V(G),E(G))$ define the distance $d(a,b)$ between two vertices $a,b\in V(G)$ to be the amount of edges in the shortest path from a to b. The similarity matrix ζ_G of the graph G is indexed by the vertices of G, and has entries

$$\zeta_G(a,b)=q^{d(a,b)}.$$

The Möbius matrix of G is then defined as $\mu_G:=\zeta_G^{-1}$. The magnitude of G is

$$\#G:=\sum_{a,b\in V(G)}\mu_G(a,b).$$

A k-path in G is a tuple of vertices $v=(v_0,v_1,\dots,v_k)$ such that $v_i\neq v_{i+1}$ for $0\le i\le k-1$. Define the length of a path v to be

$$\operatorname{len}(v)=\sum_{i=0}^{k-1}d(v_i,v_{i+1}).$$

Let $T_{k,\ell}(G)$ be the set of all k-paths of length ℓ. The (k,ℓ)-magnitude chain group of G is then

$$MC_{k,\ell}(G):=\mathbb{Z}\langle T_{k,\ell}(G)\rangle.$$

The boundary map $\partial_{k,\ell}:MC_{k,\ell}(G)\to MC_{k-1,\ell}(G)$ is defined by

$$\partial_{k,\ell}(v):=\sum_{i=1}^{k-1}(-1)^i\partial^i_{k,\ell}(v)$$

where

$$\partial^i_{k,\ell}(v):=\begin{cases}v^{\setminus i} & \text{if } \operatorname{len}(v^{\setminus i})=\ell\\ 0 & \text{otherwise.}\end{cases}$$

and $v^{\setminus i}=(v_0,\dots,v_{i-1},v_{i+1},\dots,v_k)$. Magnitude homology is then defined in the natural way

$$MH_{k,\ell}(G):=\ker(\partial_{k,\ell})/\operatorname{Im}(\partial_{k+1,\ell}).$$

Results

The Kaneta-Yoshinaga approach takes a triangulable manifold M and continues with a minimal triangulation $\widehat{M}$ as an input.Then there is a sequence of isomorphisms and and embedding

$$H_n(M)\cong H_n(\mathrm{Bd}(\widehat{M}))\cong H_{n+2}(\Delta(\widehat{M}))\cong FMH_{n+2}^{(\hat{0},\hat{1})}(\widehat{G})\hookrightarrow MH_{n+2,d(\hat{0},\hat{1})}(\widehat{G})$$

where $\mathrm{Bd}(\widehat{M})$ is the barycentric subdivision of $\widehat{M}$, $\Delta(\widehat{M})$ is the order complex of the face poset of $\widehat{M}$, and $\widehat{G}$ is the graph representing the Hasse diagram of the face poset of $\widehat{M}$. $\hat{0}$ and $\hat{1}$ are the minimal and maximal elements, respectively, of the face poset.

FMH represents the framed magnitude homology[3].

Using this technique, the homology of $\mathbb{RP}^2$ or any lens space can be embedded into the magnitude homology of some graph, yielding the desired torsion.

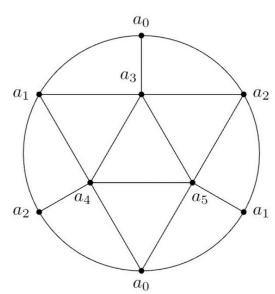

Figure 1.1: a minimal triangulation of $\mathbb{RP}^2$

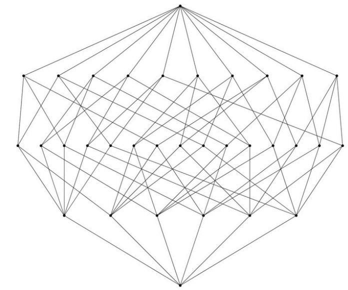

Figure 1.2: the face poset of $\mathbb{RP}^2$

We discovered is a new way to find torsion in magnitude homology of graphs, which does not come from $\mathbb{RP}^2$ or lens spaces. For an even number k and a derangement $\sigma\in S_{k-1}$ a ranked poset P_k^σ can be constructed such that it has 2-torsion in the homology of its order complex $\Delta(P_k^\sigma)$. Below are the Hasse diagrams (and consequently the graphs) for the posets $P_4^{(1\,2\,3)}$ and $P_6^{(2\,1\,3\,4\,5)}$. both of which contain fewer edges and vertices than the $\mathbb{RP}^2$ example. Computations have confirmed that both these graphs exhibit torsion in Eulerian and discriminant magnitude homology.

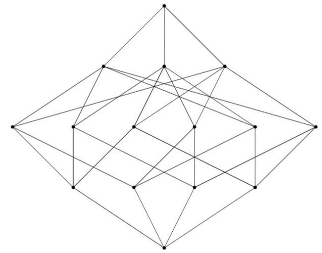

Figure 2.1: the poset $P_4^{(1\,2\,3)}$

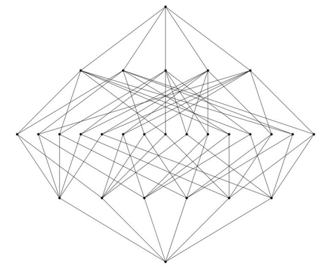

Figure 2.2: the poset $P_6^{(2\,1\,3\,4\,5)}$

Recent Developments

In early 2024, Giusti and Menara in [1] developed a theory of Eulerian and discriminant magnitude homology. Define the set of Eulerian k-paths to be

$$ET_{k,\ell}(G):=\{v\in T_{k,\ell}(G)\,|\,v_i\neq v_j,\ \forall i\neq j\}$$

so that

$$EMC_{k,\ell}(G):=\mathbb{Z}\langle ET_{k,\ell}(G)\rangle.$$

is the (k,ℓ)-Eulerian magnitude chain group of G. Since $EMC_{k,\ell}(G)\trianglelefteq MC_{k,\ell}(G)$ the boundary maps are just $\partial_{k,\ell}$ restricted to the Eulerian chain groups, so that Eulerian magnitude homology is defined in the natural way. The (k,ℓ)-discriminant magnitude chain group is the quotient

$$DMC_{k,\ell}(G):=MC_{k,\ell}(G)/EMC_{k,\ell}(G).$$

equipped with the quotient differential. Since the order in a poset is antisymmetric, it is suspected that the Kaneta-Yoshinaga sequence can be extended to

$$H_n(\Delta(P))\cong FMH_n^{(\hat{0},\hat{1})}(G)\hookrightarrow EMH_{n,d(\hat{0},\hat{1})}(G)\hookrightarrow MH_{n,d(\hat{0},\hat{1})}(G).$$

Future Directions & Acknowledgements

1. Does there exists a graph with torsion in its discriminant magnitude homology but not not in its Eulerian magnitude homology?
2. See Leinster's "100 papers on magnitude" post on n-Cat cafe.

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References

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